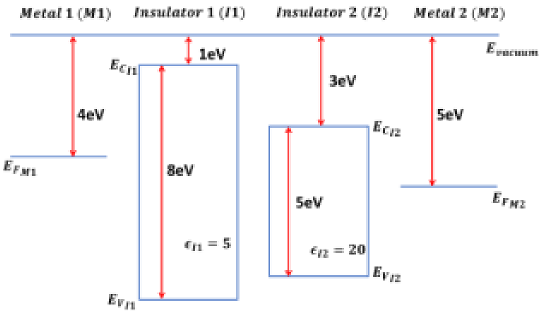
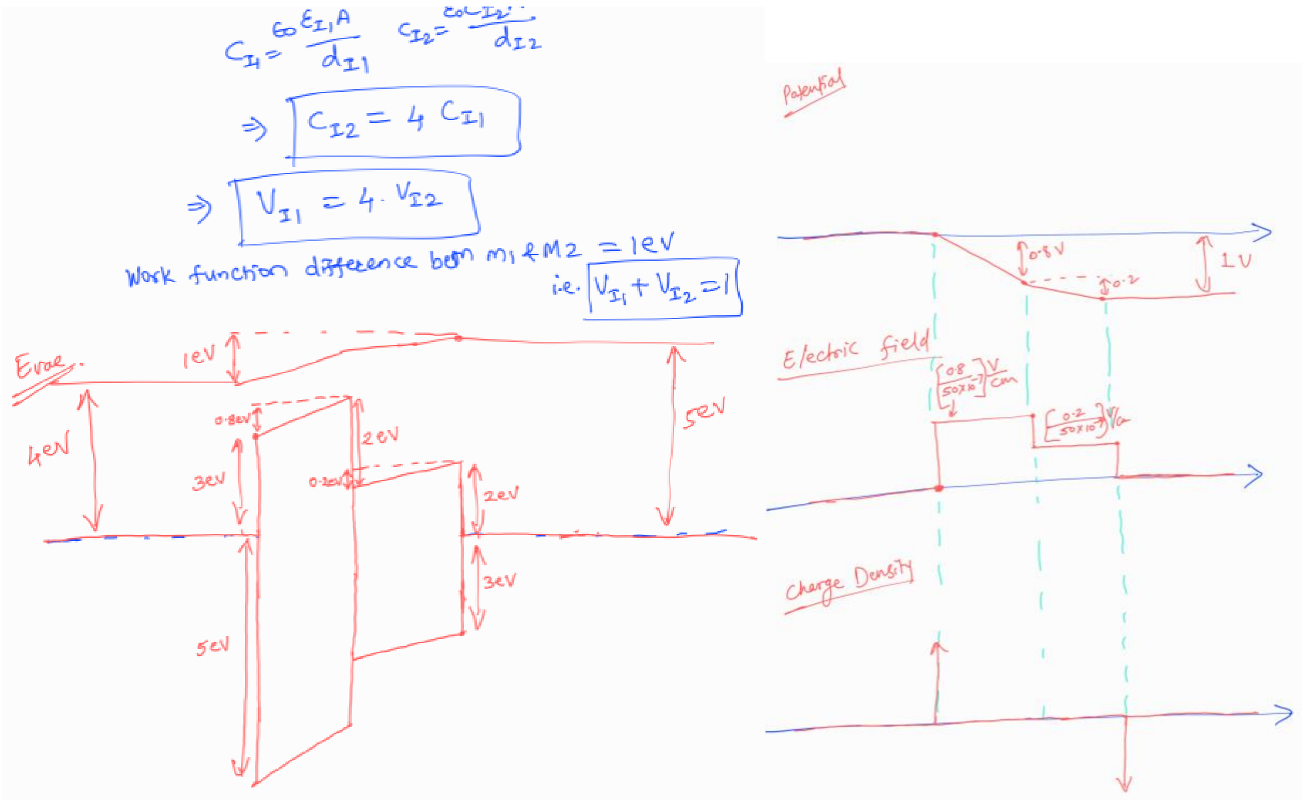
EE 226: Semiconductor Devices
Quiz – 5 Solutions

Q1. Four different materials and parameters are shown below before junction formation. Assume the length of all the materials is 50nm each. When they are connected together M1/I1/I2/M2, under thermal equilibrium - Draw the energy band diagram. Specify exact barrier heights, band bending in different regions. **(4 Marks)**



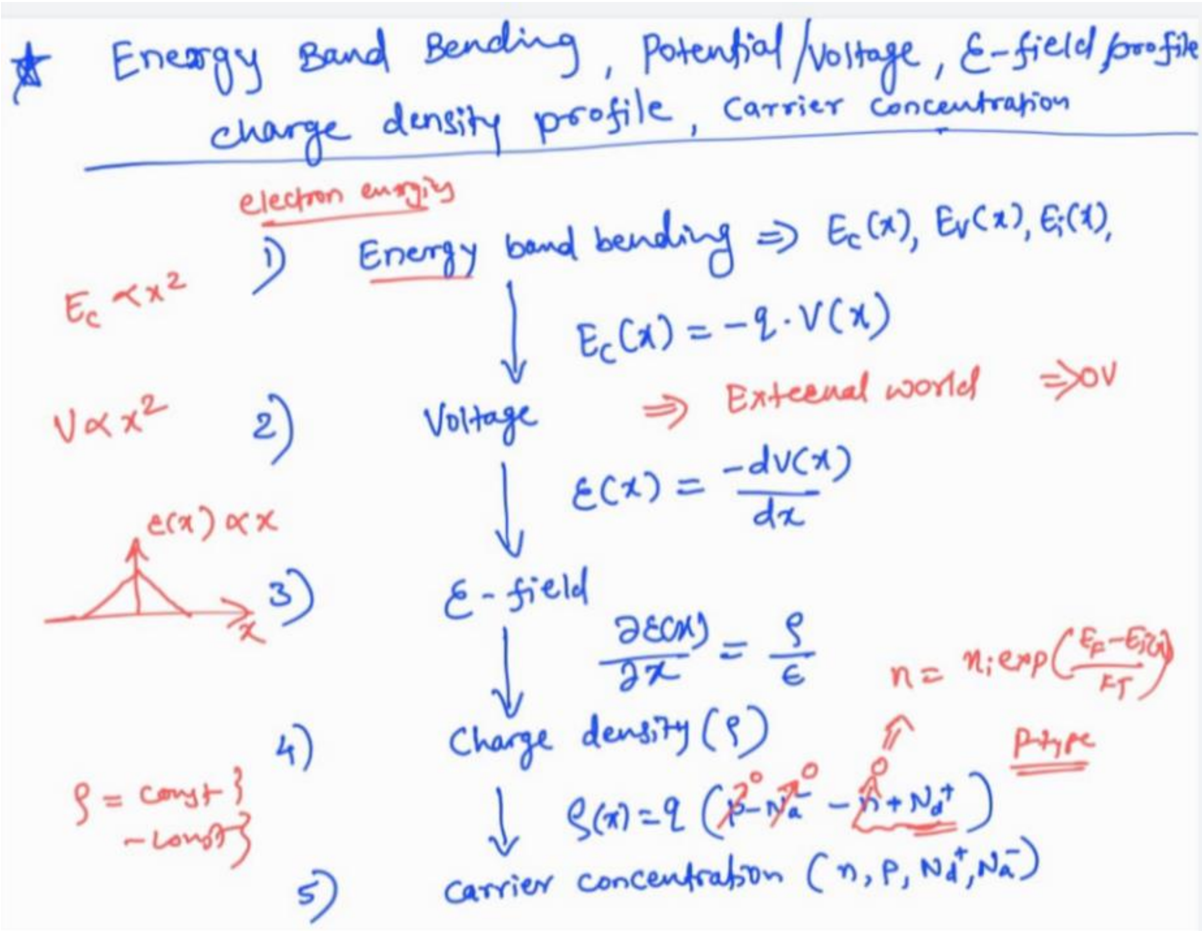
Solution:



❖ **Please Note that whoever has not put 0.8 and 0.2 eV in Band diagram will get a deduction of 1 marks.**

Q2. With a flowchart, write the relationship between band diagram, Potential, Electric Field, charge density and carrier concentration. Specify units for all. **(3 Marks)**

Solution:



Units

Band Diagram is represented with units of eV.

Potential is represented with units of V.

Electric field is represented with units of V/cm.

Charge Density is represented with units of coulomb per cubic meter (C/cm^3).

Carrier concentration is represented with units of per cubic centimetre ($/cm^3$).

Q3. Draw the approximate Charge density, Electric Field, Potential profile and Band Diagram for PN junction diode. **(3 Marks)**

Solution:

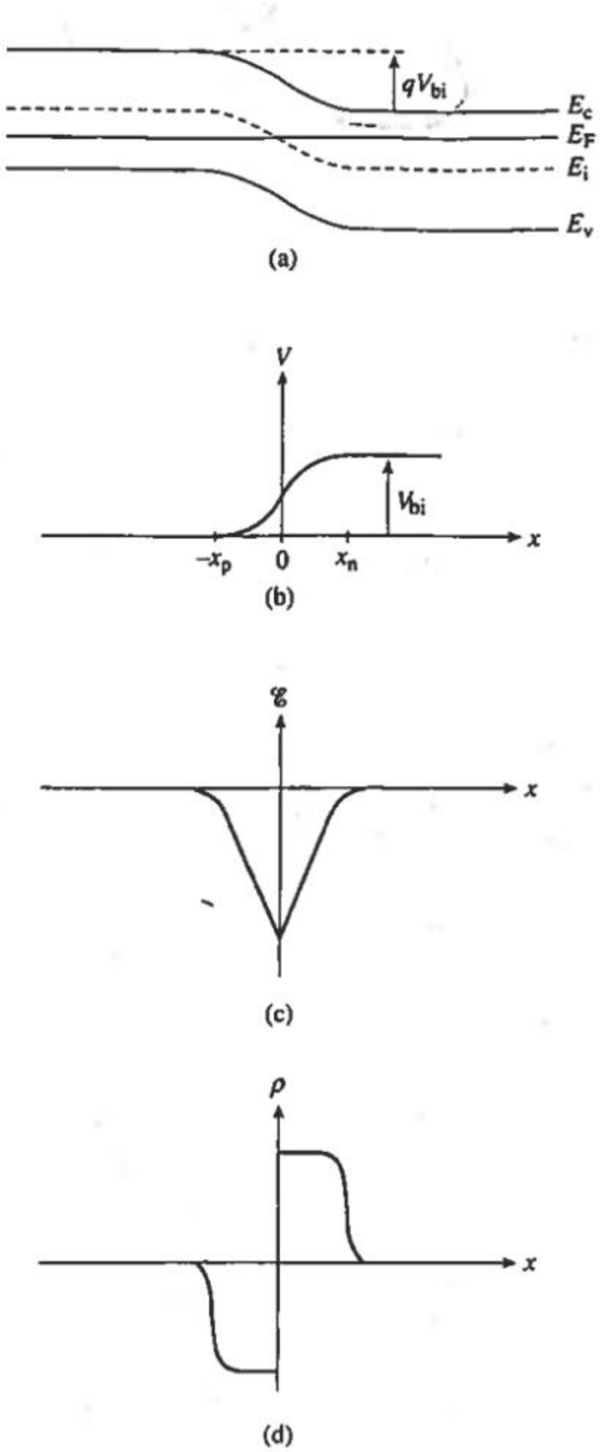


Figure 5.4 General functional form of the electrostatic variables in a *pn* junction under equilibrium conditions. (a) Equilibrium energy band diagram. (b) Electrostatic potential, (c) electric field, and (d) charge density as a function of position.

- ❖ **Please Note box like figure for Charge density is also acceptable. Also no curve at -x_p and x_n is also acceptable.**